

Exercise: Suppose that $\mathbf{X}$ is multivariate normal with

$$\boldsymbol{\mu} = \begin{bmatrix} 4 \\ 6 \\ 3 \end{bmatrix}$$

and

$$\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 0.5 & 0 \\ 0.5 & 2 & 0.1 \\ 0 & 0.1 & 3 \end{bmatrix}.$$

- 1 What is the distribution for (X_1, X_3) ? Which pairs of random variables are
 2 independent, if any?

(X_1, X_3) is bivariate normal with

$$\boldsymbol{\mu} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

and $\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$

We see that $\text{Cov}(X_1, X_3) = 0$, so X_1, X_3
 are independent.

But: X_1, X_2 are dependent
 X_2, X_3 " "

Example: Portfolio Theory

Let R_i denote the return on equity i for $i = 1, 2, \dots, n$, and let w_i denote the weight placed on equity i in the portfolio. Assume that $\sum_i w_i = 1$ and $w_i \geq 0$.

Let the overall return be $Y = \sum_i w_i R_i = \underline{w}^T \underline{R}$

Exercise: Describe the distribution of Y , and what properties an optimal portfolio would possess.

$$E(Y) = \underline{w}^T E(\underline{R})$$

want this to be large

$$V(Y) = \underline{w}^T V(\underline{R}) \underline{w}$$

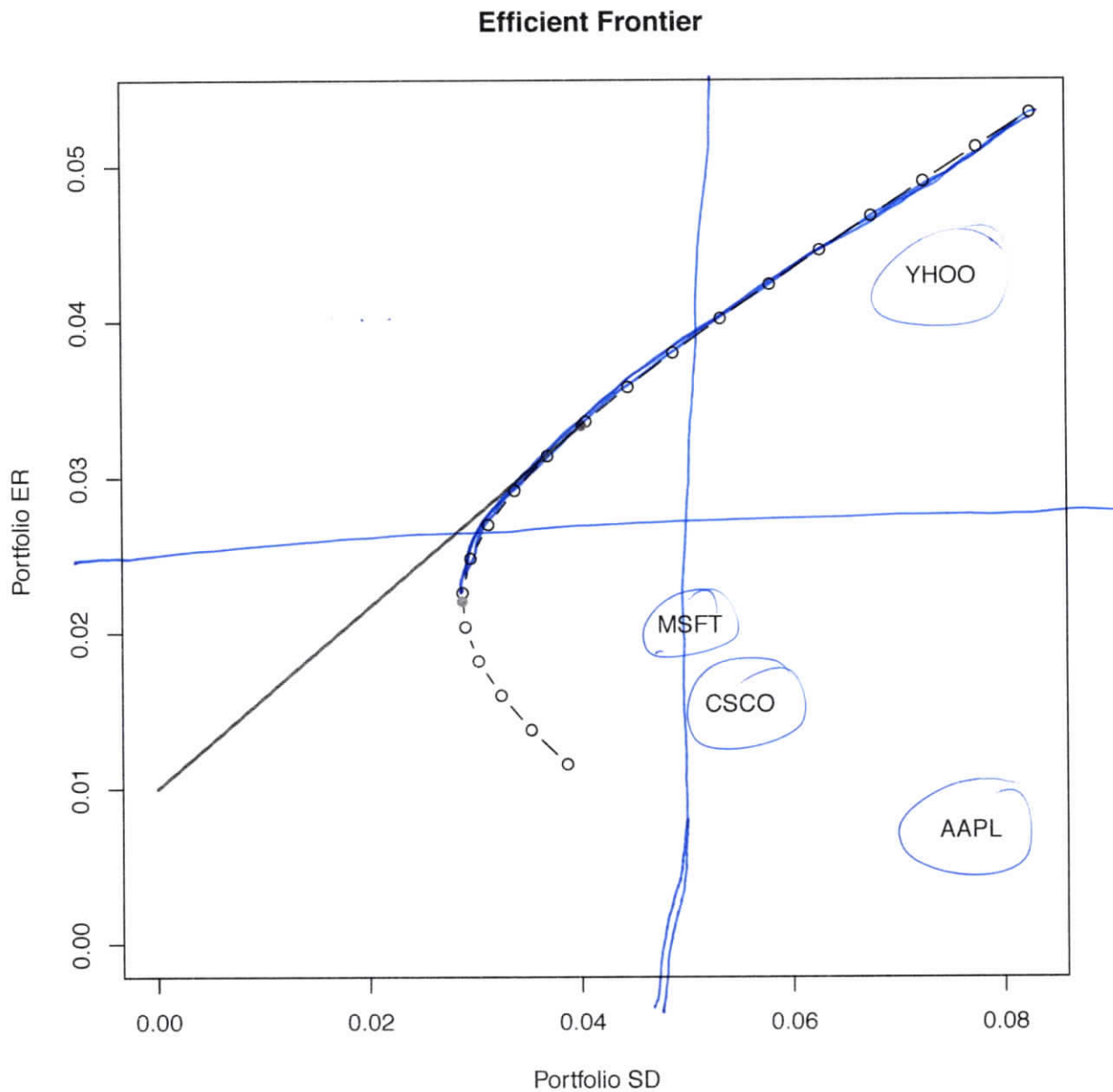
want this to be small

Any choice of $\underline{w}$ lands somewhere in the (SD, EV) plane

Among strategies with the same EV, prefer one with smallest SD

Among those with the same SD, prefer one with largest EV

This curves out the efficient frontier



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2 Based on monthly returns from September 1, 2012 to September 1, 2014.

Part 4: Conditional Expectation

In this Part we will present some useful results concerning **conditional expectation**.

The simplest example of conditional expectation would likely be the case of a discrete random variable X , when conditioning on an event B , i.e.,

$$E(X|B) = \sum_k kP(X = k|B).$$

It is often the case that the event B is an event concerning X itself, e.g.,

$$E(X|X \geq k).$$

The conditional distributions we derive in Part 3 play a key role here. We can naturally define

$$E(Y|X = x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy$$

$$E(Y) = \int_{-\infty}^{\infty} y f_Y(y) dy$$

or

$$E(Y|X = x) = \sum_y y f_{Y|X}(y|x)$$

in the cases where Y is continuous and discrete, respectively.

Exercise: Suppose that the joint pdf of (X, Y) is given by:

$$f_{XY}(x, y) = \begin{cases} 24xy & x > 0, y > 0 \\ & x + y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

What is the expected value of Y given that X equals x ?

$$f_{Y|X}(y|x) = \begin{cases} \frac{24xy}{12x(1-x)^2}, & 0 < x < 1 \\ & 0 < y < 1-x \\ 0 & , \text{ otherwise} \end{cases}$$

$$E(Y|X=x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy$$

$$= \int_0^{1-x} \frac{2y^2}{(1-x)^2} dy, \quad 0 < x < 1$$

$$= \frac{2}{3}(1-x), \quad 0 < x < 1$$

$$E(Y) = \frac{2}{5} \neq E(Y|X=x)$$

Laws of Total Probability

Assume that events $B_1, B_2, \dots$ form a partition of Ω , and that A is another event. Then each of the following hold.

$$P(A) = \sum_i P(A|B_i)P(B_i)$$

$$E(X) = \sum_i E(X|B_i)P(B_i)$$

$$P(A|C) = \sum_i P(A|C, \underline{B_i})P(B_i|C)$$

$$E(X|C) = \sum_i E(X|C, B_i)P(B_i|C)$$

$$f_X(x) = \sum_y f_{X|Y}(x|y) f_Y(y) \quad \text{if } Y \text{ discrete}$$

$$\underline{f_X(x)} = \int f_{X|Y}(x|y) f_Y(y) dy \quad \text{if } Y \text{ continuous}$$

$$= \int f_{X,Y}(x,y) dY$$

Exercise (From HOTS): What is the expected number of tosses of an unfair coin needed to get two heads in a row (assume probability p of a head)?

$X = \#$ of tosses required to see two heads in row
 $Y_i = \text{outcome of } i^{\text{th}} \text{ toss } (H, T)$

$$\begin{aligned} E(X) &= E(X | Y_1 = H) P(Y_1 = H) + E(X | Y_1 = T) P(Y_1 = T) \\ &= \underbrace{E(X | Y_1 = H)} p + (1 + E(X)) (1 - p) \end{aligned}$$

$$\begin{aligned} E(X | Y_1 = H) &= E(X | Y_1 = H, Y_2 = H) P(Y_2 = H | Y_1 = H) \\ &\quad + E(X | Y_1 = H, Y_2 = T) P(Y_2 = T | Y_1 = H) \\ &= (2) p + (2 + E(X)) (1 - p) \end{aligned}$$

$$= E(X) (1 - p^2) + p + 1$$

Solve for $E(X)$, find

$$E(X) = \frac{p+1}{p^2}$$

Exercise (From HOTS): I will roll a single die no more than three times. You can stop me immediately after the first roll, or immediately after the second, or you can wait for the third. I will pay you the same number of dollars as there are dots on the single upturned face on my last roll (roll number three unless you stop me sooner). What is your playing strategy?

Consider the " (a, b) strategy": Stop after roll 1 if it's at least a , stop after roll 2 if it's at least b .

Let $X = \text{winning}$

$A_i = \{\text{stop after toss } i\}$

$Y_i = \text{outcome of toss } i$

Fix a, b

$$E(X) = E(X|A_1)P(A_1) + E(X|A_2)P(A_2) + E(X|A_3)P(A_3)$$

$$E(X|A_1) = E(Y_1 | Y_1 \geq a) = \sum_{k=a}^6 k P(Y_1 = k | Y_1 \geq a)$$

$$= \sum_{k=a}^6 k \frac{P(Y_1 = k, Y_1 \geq a)}{P(Y_1 \geq a)}$$

$$= \sum_{k=a}^6 k \frac{\left(\frac{1}{6}\right)}{\left(\frac{7-a}{6}\right)} = \frac{1}{7-a} \left(21 - \frac{a(a-1)}{2} \right)$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$E(X|A_2) = \left(\frac{1}{7-b}\right) \left(21 - \frac{b(b-1)}{2}\right)$$

$$E(X|A_3) = 3.5$$

$$P(A_1) = P(Y_1 \geq a) = \frac{7-a}{6}$$

$$P(A_2) = P(Y_1 < a, Y_2 \geq b) = \left(\frac{a-1}{6}\right) \left(\frac{7-b}{6}\right)$$

since tosses are independent

$$P(A_3) = P(Y_1 < a, Y_2 < b) = \left(\frac{a-1}{6}\right) \left(\frac{b-1}{6}\right)$$

b

a

	[, 1]	[, 2]	[, 3]	[, 4]	[, 5]	[, 6]
[1,]	3.500000	3.500000	3.500000	3.500000	3.500000	3.500000
[2,]	3.916667	3.986111	4.027778	4.041667	4.027778	3.986111
[3,]	4.166667	4.305556	4.388889	4.416667	4.388889	4.305556
[4,]	4.250000	4.458333	4.583333	4.625000	4.583333	4.458333
[5,]	4.166667	4.444444	4.611111	4.666667	4.611111	4.444444
[6,]	3.916667	4.263889	4.472222	4.541667	4.472222	4.263889

$$(5)(0.5) + (3.5)(0.5) = 4.25$$

Exercise (From HOTS): A coin-making machine produces pennies. Each penny is manufactured to have a probability P of turning up heads. However, the machine draws P randomly from the uniform distribution on $[0, 1]$ so P can differ for each coin produced. A coin pops out of the machine. You flip it once, and it comes up heads. Given this information, what is the (conditional) cumulative distribution function $F_{P|H}(p)$ for the probability of a head for that coin (where "H" denotes conditioning on the head)?

What is the (conditional) distribution for the probability of a head if you flip the coin 1,000 times and get 750 heads?

Assume make n flips. $X = \#$ of heads, $0 \leq P \leq 1$
Find $P(P \leq p | X = x)$

Note that
$$P(P \leq p | X = x) = \int_0^p f_{P|X}(u|x) du$$

$$= \int_0^p \left[\frac{f_{X|P}(x|u) f_P(u)}{\int_{-\infty}^{\infty} f_{X|P}(x|v) f_P(v) dv} \right] du$$

since
$$f_{P|X} = \frac{f_{P,X}}{f_X} = \frac{f_{X|P} f_P}{\int_{-\infty}^{\infty} f_{X|P}(x|v) f_P(v) dv}$$

know $f_P(p) = 1$, $0 \leq p \leq 1$

$$f_{X|P}(x|p) = \binom{n}{x} p^x (1-p)^{n-x} \quad x=0,1,\dots,n$$

$$\text{So, } \int_0^p \binom{n}{x} u^x (1-u)^{n-x} (1) du$$

$$= \frac{\int_0^p \binom{n}{x} u^x (1-u)^{n-x} (1) du}{\int_0^1 \binom{n}{x} v^x (1-v)^{n-x} (1) dv}$$

$$= \int_0^p \left(\frac{\Gamma(n+2)}{\Gamma(x+1)\Gamma(n-x+1)} \right) u^x (1-u)^{n-x} du$$

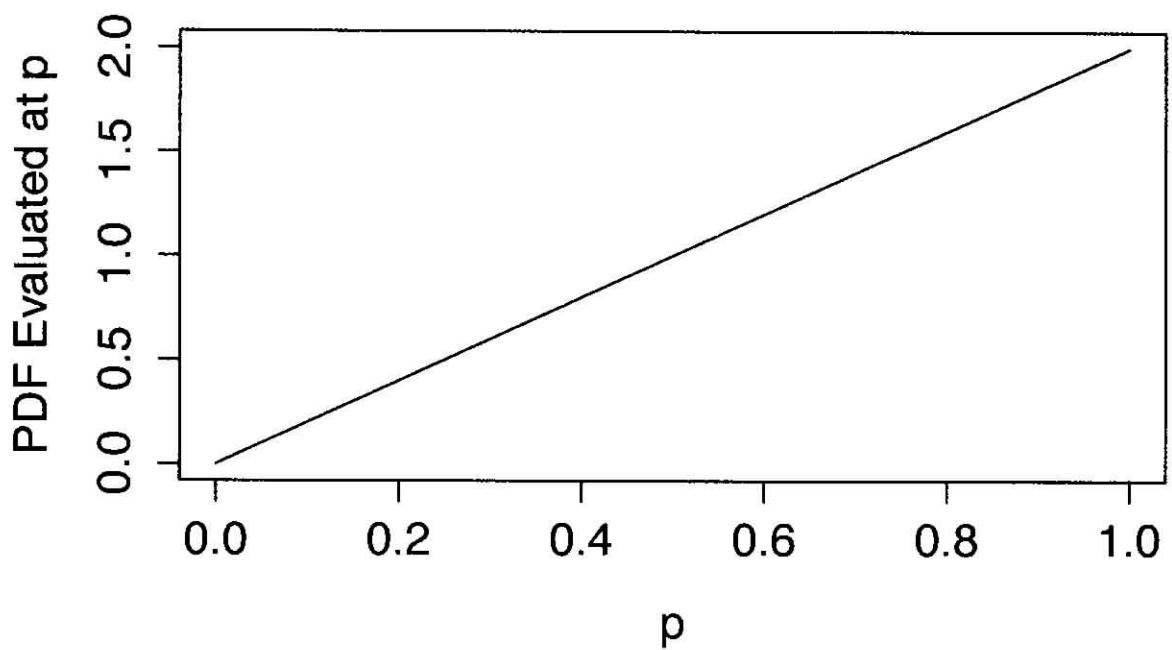
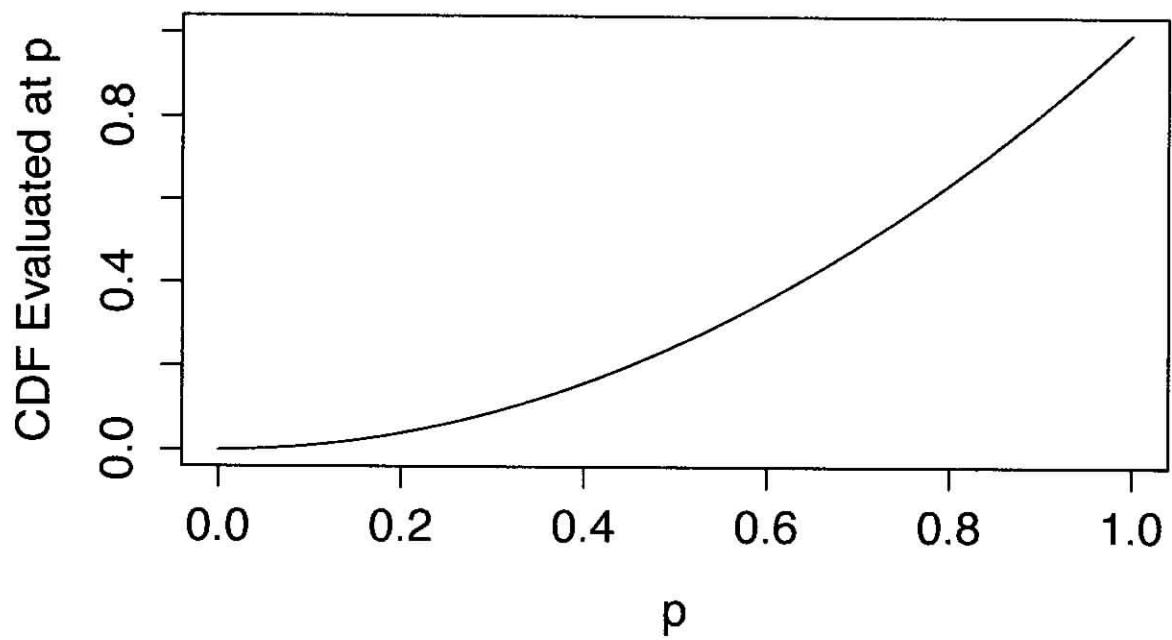
Math Facts

$$= P(U \leq p) \quad U \sim \text{Beta}(x+1, n-x+1)$$

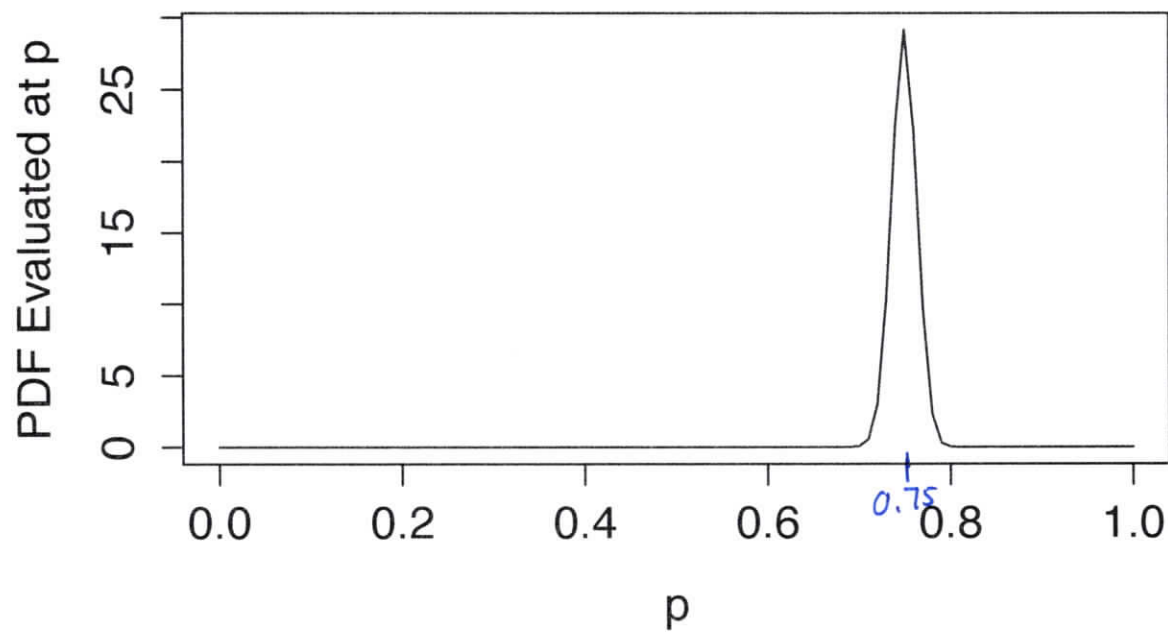
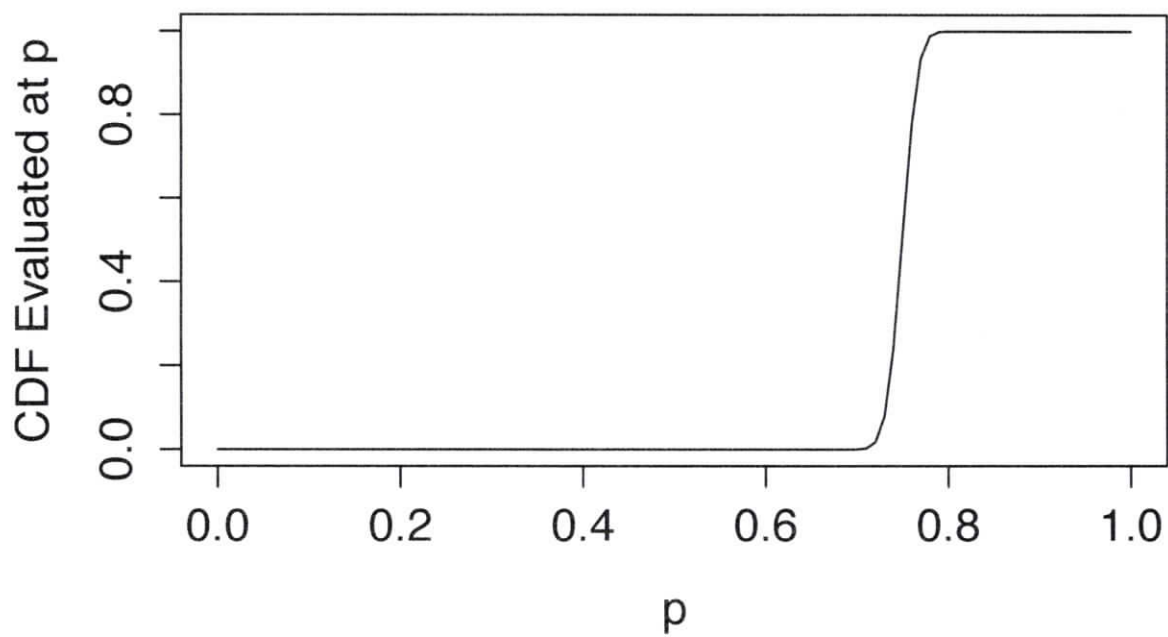
Given x heads in n tosses, P has the $\text{Beta}(x+1, n-x+1)$ dist.

If $n=1$, $x=1$, P is $\text{Beta}(2,1)$
 CDF is p^2 for $0 \leq p \leq 1$

- For the case where $n = 1$ and $x = 1$:



1 For the case where $n = 1000$ and $x = 750$:



2

Iterated Conditioning

There is convenient shorthand for the above version of the Law for Total Probability for Expectations. We often write

$$E(X) = E(E(X|Y))$$

This expression works for both discrete and continuous random variables.

Compare to
$$E(X) = \int_{-\infty}^{\infty} E(X|Y=y) \underline{f_Y(y)} dy$$

It also holds that

$$V(X) = V(E(X|Y)) + E(V(X|Y))$$

Exercise: Explain the logic behind the iterated conditioning formula.

Note that $E(X|Y)$ is a RV, a function on Ω

Let $Z = E(X|Y)$, $Z(\omega) = E(X|Y=Y(\omega))$

ie. Z captures all the possible values of Y we could condition on for outcome $\omega_1 \in \Omega$, $Y=Y(\omega_1)$

So, $E(Z) = E(E(X|Y))$ takes a weighted average of all possible ω (all possible $Y(\omega)$) with weighting provided by f_Y

Exercise: The lifetime of a part can be modelled using an Exponential(λ) distribution. These parts are produced in lots, and it is most realistic to assume that λ varies from lot to lot. Based on experience, the manufacturer models the lot-to-lot variability in λ using the pdf

$$f_{\Lambda}(\lambda) = 4\lambda e^{-2\lambda}, \quad \lambda > 0 \quad \Lambda \sim \text{Gamma}(2, 2)$$

Determine the unconditional probability density function for the length of life of a part. What is the probability that a randomly chosen part lasts longer than 2 time units? What is the expected length of the life of a part?

$X = \text{length of life of part}$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|\Lambda}(x|\lambda) f_{\Lambda}(\lambda) d\lambda$$

$$= \int_0^{\infty} \lambda e^{-\lambda x} 4\lambda e^{-2\lambda} d\lambda, \quad x > 0$$

$$= \frac{8}{(x+2)^3}, \quad x > 0 \quad \text{see Math Facts}$$

$$P(X > 2) = \int_2^{\infty} \frac{8}{(x+2)^3} dx$$

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx = \dots$$

Or :

$$E(X) = E(E(X | \Lambda))$$
$$= E\left(\frac{1}{\Lambda}\right)$$

$$= \int_0^{\infty} \frac{1}{\lambda} 4\lambda e^{-2\lambda} d\lambda$$

Rule
for Lazy
Stat.

$$= 2$$

- 1 **Exercise:** Consider a Poisson process with rate λ . Suppose that "events"
 2 are of two types: "Type I" and "Type II." An event is of Type I with proba-
 3 bility p , and the type is determined independently from event to event. Let
 4 X denote the number of events in one time unit, so that X is $\text{Poisson}(\lambda)$.
 5 Let Y denote the number of events in one time unit that are of Type I. Find
 6 $E(Y)$ and $V(Y)$.

$$E(Y) = E(E(Y|X))$$

$$= E(Xp) \quad Y|X \sim \text{binomial}(X, p)$$

$$= p E(X) = p\lambda \quad \text{since } X \sim \text{Poisson}(\lambda)$$

$$V(Y) = V(E(Y|X)) + E(V(Y|X))$$

$$= V(Xp) + E(Xp(1-p))$$

$$= p^2 V(X) + p(1-p) E(X)$$

$$= p^2 \lambda + p(1-p) \lambda$$

$$= \lambda p$$

22 Is $Y \sim \text{Poisson}(\lambda p)$? Yes, ^{but} we have not yetst proven that.

$$V(Y) = E(Y^2) - E(Y)^2$$

$$= E(E(Y^2|X)) - E(E(Y|X))^2$$

$$= E(V(Y|X) + E(Y|X)^2) - E(E(Y|X))^2$$

$$E(Z^2) = V(Z) + E(Z)^2$$

$$= E(V(Y|X)) + E(E(Y|X)^2) - E(E(Y|X))^2$$

$$= E(V(Y|X) + V(E(Y|X)))$$